4th

Monday, August 31, 2026 9:53 AM

2 pt BVP

$$\begin{cases} -(au')' = f \quad \text{on } (0,1) \\ u(0) = u(1) = 0 \end{cases}$$

↓ variational problem

Find $u \in V$ s.t

$$\int au'v' \, dx = \int_0^1 fv \, dx \qquad \forall\, v \in V$$

where, $V = \left\{ v : \int_0^1 v^2 \, dx < \infty, \right.$

$$\left. \int_0^1 (v')^2 \, dx < \infty \right.$$

$$v(0) = v(1) = 0$$

mesh T_h : mesh with grid points

$x_0 = 0$, x_1 x_2, ... x_m $x_{m+1} = 1$

FEM $V_h := \left\{ v \in V : v \Big|_{x_i, x_{i+1}} \in \text{polynomial of degree } 1 \quad \forall\, i = 0, \dots, m \right\}$

FEM: find $U \in V_h \subset V$

s.t $\int_0^1 a u' v' dx = \int_0^1 f v dx$

$\forall \quad v \in V_h$

A basis for V_h:

$\phi(x), \phi_2(x), \phi_3(x), \cdots\cdots, \phi_m(x)$

where, $\phi_j(x) = \begin{cases} \dfrac{x - x_{j-1}}{h_j} & x_{j-1} < x < x_j \\ \dfrac{x_{j+1} - x_j}{h_{j+1}} & x_j < x < x_{j+1} \end{cases}$

$V_h = \text{Span}\{\phi_1, \phi_2, \cdots\cdots \phi_m\}$

$\dim V_h = m$

if $U \in V_h$, then

$$U(x) = \sum_{i=1}^{m} \xi_j \phi_j(x)$$

where, $\xi_j = U(x_j)$

for $j = 1, 2, \cdots, m$.

Galerkin Orthogonality:

$$\int_0^1 a u'v' dx = \int_0^1 f v dx \rightarrow u, v \in V$$

Then FEM, $\int_0^1 a U'v' dx = \int_0^1 f v dx \rightarrow U, v \in V_h$

Subtract:

$$\int_0^1 a(u-U)'v' dx = 0 \quad \text{for all } v \in V_h.$$

Let, $U(x) = \sum_{j=1}^{m} \xi_j \phi_j(x)$

$v = \phi_i, \quad i = 1, 2, 3, \cdots m$

$\int_0^1 r \simeq 0 \cdot v' \phi_i' dx = \int_0^1 f \phi_i dx$

$$\int_0^1 a \left(\sum_{j=1}^{m} \xi_j \phi_j \right)' \phi_i' \, dx = \int_0^1 f \phi_i \, dx$$

$$= \sum_{j=1}^{m} \xi_j \int_0^1 a \phi_i' \phi_j' \, dx = \int_0^1 f \phi_i \, dx$$

This is an $m \times m$ linear system:

$$\underline{\underline{A}} \, \underline{\xi} = \underline{b}$$

where, $A_{ij} = \int_0^1 a \phi_i' \phi_j' \, dx$

and $b_i = \int_0^1 f \phi_i \, dx$

$$\underline{\xi} = (\xi_1, \xi_2, \xi_3, \cdots \xi_m)$$

$\underline{\underline{A}}$: stiffness matrix

(1) symmetric : test and trial function are from the same space

(2) positive definitive $a > 0$

(4) positive definitive $a > 0$

(3) tridiagonal : basis function have compact support on $[x_{i-1}, x_{i+1}]$

$$\int_0^1 a \phi_i' \phi_j' \, dx \neq 0$$

for $j = i-1, i, i+1$

ex: $a \equiv 1$ and take $h_j \equiv h \ \forall j$ ← uniform mesh.

$$A_{ii} = \int_0^1 1 \cdot \phi_i' \phi_i' \, dx$$

$$= \int_{x_{i-1}}^{x_i} (\phi_i')^2 \, dx + \int_{x_i}^{x_{i+1}} (\phi_i')^2 \, dx$$

$$= \frac{1}{h^2}(x_i - x_{i-1}) + \frac{1}{h^2}(x_{i+1} - x_i)$$

$$= \frac{2}{}$$

$$= \frac{2}{h}$$

$$A_{(i,i+1)} = \int_{x_i}^{x_{i+1}} \phi_i' \, \phi_{i+1}' \, dx = -\frac{1}{h}$$



$$b_i = \int_{x_{i-1}}^{x_{i+1}} f(x)\, \phi_i(x)\, dx$$

Composite trapezoidal rule $\approx \frac{h}{2}\left[f(x_{i-1})\,\phi_i(x_{i-1}) + 2f(x_i)\,\phi_i(x_i) + f(x_{i+1})\,\phi_i(x_{i+1})\right]$ ($\phi_i(x_{i-1}) = 0$, $\phi_i(x_i) = 1$, $\phi_i(x_{i+1}) = 0$)

$$= h f(x_i)$$

$$\underline{\underline{A}} = \frac{1}{h}\begin{bmatrix} 2 & -1 & & & \\ -1 & 2 & -1 & & \\ & -1 & 2 & -1 & \\ & & \ddots & \ddots & \ddots \\ & & & -1 & 2 & -1 \end{bmatrix}$$

$$b = h\left(f(x_1), f(x_2), \ldots\ldots f(x_m)\right)$$

FDM will also provide the same Mathematic

FDM will also provide the same Mathematic as well as FEM ~~for~~ 2-P BVP in FDM

Note: this is exactly $O(h^2)$ central Finite difference method.